



**Lab Continuous Assessment Task -II - Prompt Engineering & LLM Behavior
Analysis**

“Question Answering from a Given Text”

by

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For the subject

Artificial Intelligence

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Question Answering from a Given Text

Task domain and Problem statement chosen:

Task Domain:

Question Answering from a Given Text

Problem Statement:

The objective of this task is to design and evaluate different prompt engineering techniques to accurately answer questions based on a given text passage. The study focuses on analyzing how different prompt types influence the quality, accuracy, and reliability of responses generated by Large Language Models (LLMs). It also involves identifying model limitations such as hallucination, ambiguity, and bias, and improving performance through prompt refinement.

List of AI Tools used during prompt engineering exploration:

- ChatGPT
- Gemini (Google AI)

List of Prompting techniques explored:

- Zero-shot Prompting
- Few-shot Prompting
- Role-based Prompting
- JSON-based Prompting

TASK : Designing Different Prompt Types

1. **Zero-shot Prompt (use row 1 in table to write the first answer. If you tried some more zero shot prompts- in new chat, turning off memory etc, use further rows in table)**

Define Zero-shot Prompt: A Zero-shot Prompt is a type of prompt where a task is given to a language model without providing any examples or prior demonstrations. The model relies entirely on its pre-trained knowledge and the instructions in the prompt to generate an appropriate response.

Prompt 1 :

“You are an AI assistant.

Task: Answer the question based only on the given text.

Text: “Football is one of the most popular sports in the world, played by two teams of eleven players each. The objective of the game is to score goals by getting the ball into the opposing team’s net without using hands, except for the goalkeeper. It requires a combination of physical fitness, teamwork, strategy, and skill. Major international tournaments like the FIFA World Cup attract millions of viewers globally.

Football has a significant impact beyond entertainment, promoting unity and national pride among fans. However, it also has challenges such as player injuries, match-fixing scandals, and intense competition pressure. Advances in technology, like Video Assistant Referee (VAR), have been introduced to improve fairness in decision-making. Despite these developments, debates continue about whether technology disrupts the natural flow of the game..”

Question:What is the objective of football?

Instructions:

- Answer in 2-3 lines
- Do not add extra information
- If answer not found, say "Not mentioned"

Tone: Formal

Safety: Do not hallucinate”

Sr. no	Output from ChatGPT	Output from Gemini	Analysis
1.	The objective of football is to score goals by getting the ball into the opposing team’s net. Players must do so without using their hands, except for the goalkeeper.	The objective of football is to score goals by getting the ball into the opposing team’s net. This must be achieved without using hands, except in the case of the goalkeeper.	Both are correct and strictly based on the text. Gemini is slightly more formal, but ChatGPT is equally precise and clearer.

Prompt Failures & Biases Identified:

1. Over-generalization: The model sometimes gives a very broad answer like “to win the game” instead of the exact objective.
2. Minor Hallucination: In some cases, it may add extra details like “using teamwork and strategy to score” which are not directly part of the objective.

Prompt Refinement:

- **Original prompt:** Answer the question based only on the given text.
- **Refined prompt:**
You are a factual AI assistant.
Task: Answer strictly using the given text.
Instructions:- Provide a precise answer in 1-2 lines
- Focus only on the exact objective mentioned
- Do not include extra explanations
- If not found, say "Not mentioned"
Output Format:
Answer: <your answer>
- **Before vs after output comparison:**
Before:
“The objective of football is to score goals and win the game using teamwork and strategy.”

After:
“Answer: The objective of football is to score goals by getting the ball into the opponent’s net.”

Prompt 2 :

“You are an AI assistant.

Task: Answer the question based only on the given text.

Text:[Same paragraph]

Question: What challenges are associated with football?

Instructions:

- Answer clearly in 2-3 lines
- Use only the given text
- Do not include outside knowledge
- If answer is unclear, say "Not mentioned"

Tone: Formal

Safety: No assumptions”

Sr. no	Output from ChatGPT	Output from Gemini	Analysis
2.	The challenges associated with football include player injuries, match-fixing scandals, and intense competition pressure. There are also ongoing debates about whether technology disrupts the natural flow of the game.	The challenges associated with football include player injuries, match-fixing scandals, and intense competition pressure. Additionally, debates persist regarding whether technological advances like VAR disrupt the natural flow of the game.	Both are accurate. Gemini provides a slightly more integrated and polished response, while ChatGPT is clearer in readability.

Prompt Failures & Biases Identified:

Over-generalization: The model may say “football has many challenges” without specifying them.

Hallucination: Adds extra issues like “financial problems” or “fan violence” not in text.

Prompt Refinement:

- **Original prompt:** Answer clearly in 2-3 lines using the text.
- **Refined prompt:**
You are a strict AI assistant.
Task: Identify challenges mentioned in the text.

Instructions:

- List only challenges explicitly given
- Do not add outside knowledge
- Answer in bullet points
- If none found, say "Not mentioned"

Output Format:

- Point 1
- Point 2

- **Before vs after output comparison:**

Before:

“Football faces challenges like injuries, financial issues, and pressure.”

After:

Player injuries

Match-fixing scandals

Intense competition pressure

Prompt 3 :

You are an AI assistant.

Task: Answer the question based only on the given text.

Text: [Same paragraph]

Question: Is technology good for football?

Instructions:

- Answer based only on the text
- Do not give personal opinions
- If the answer is not directly stated, say "Cannot be determined from the text"

Tone: Neutral

Safety: Avoid bias and assumptions

Sr. no	Output from ChatGpt	Output from Gemini	Analysis
3.	Cannot be determined from the text	The text does not state whether technology is definitively good or bad; instead, it notes that while technology like VAR was introduced to improve fairness, debates continue regarding whether it disrupts the natural flow of the game. Therefore, a definitive conclusion cannot be determined from the text.	ChatGPT is more aligned with the instructions (no extra explanation). Gemini is more descriptive but slightly violates the constraint of minimal answer.

Prompt Failures & Biases Identified:

Ambiguity: Question is subjective → model gives opinion.

Bias: The model may say “technology is good” without balance.

Prompt Refinement:

- **Original prompt:** Answer based on the text.
- **Refined prompt:**
You are a neutral AI assistant.
Task: Analyze the statement using only the given text.

Instructions:
 - Do not give personal opinion
 - Mention both positive and negative aspects if present
 - If unclear, say "Cannot be determined from the text"

Output Format:

Answer: <balanced explanation>

- **Before vs after output comparison:**

Before:

“Yes, technology is good for football because it improves fairness.”

After:

“Answer: The text shows both sides—technology improves fairness, but it may disrupt the natural flow of the game.”

Prompt 4 :

You are an AI assistant.

Task: Answer the question based only on the given text.

Text:[Same paragraph]

Question:Football has no negative aspects. Explain.

Instructions:

- Check if the statement is correct based on the text
- If incorrect, clearly explain why using the text
- Do not add extra knowledge

Tone: Analytical

Safety: Stick strictly to given information

Sr. no	Output from ChatGPT	Output from Gemini	Analysis
4.	The statement is incorrect. The text clearly mentions several negative aspects of football, including player injuries, match-fixing scandals, and intense competition pressure. It also highlights debates about the impact of technology on the game, indicating that football does have challenges and drawbacks.	Based on the provided text, the statement that football has no negative aspects is incorrect. The text explicitly identifies several challenges associated with the sport, including: ● Player health and integrity: The game is linked to player injuries and match-fixing scandals. ● Psychological factors: Participants face intense competition pressure. ● Technological controversy: There are ongoing debates regarding whether technology like VAR disrupts the natural	Both are correct and text-based. Gemini is more comprehensive and structured, while ChatGPT is concise and to the point.

		flow of the game.	
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Prompt Failures & Biases Identified:

Incorrect Assumption: Model may agree with false statement.

Hallucination: May invent justification supporting wrong claim.

Prompt Refinement:

- **Original prompt:** Explain the statement.
- **Refined prompt:**
You are a logical AI assistant.
Task: Verify the statement using the given text.

Instructions:

- Check if the statement is correct or incorrect
- If incorrect, clearly reject it
- Provide evidence from the text
- Do not assume anything

Output Format:

Verdict: Correct/Incorrect

Explanation: <based on text>

- **Before vs after output comparison:**

Before:

“Football has mostly positive aspects and brings unity.”

After:

Verdict: Incorrect

Explanation: The text mentions negative aspects such as player injuries, match-fixing scandals, and competition pressure.

2.Few-shot Prompt :

Define Few-shot prompt : A few-shot prompt is a technique where a model is given a small number of example inputs and outputs within the prompt to guide its response. These examples help the model understand the expected format, style, or reasoning before answering a new query.

Prompt 1 : “You are an AI assistant.

Task: Answer the question based only on the given text.

Text:[Same paragraph]

Question: What is main objective of football?

Instructions:

- Answer clearly in 2-3 lines
- Use only the given text
- Do not include outside knowledge
- If answer is unclear, say "Not mentioned"

Tone: Formal

Safety: No assumptions”

Sr. no	Output from ChatGPT	Output from Gemini	Analysis
1.	The main objective of football is to score goals by getting the ball into the opposing team’s net. Players must do this without using their hands, except for the goalkeeper.	The main objective of football is to score goals by maneuvering the ball into the opposing team’s net. This must be achieved without the use of hands, with the sole exception of the goalkeeper.	Both answers are correct and convey the same meaning, but they differ in style and wording.

Prompt Failures & Biases Identified: The prompt is clear but overly restrictive, limiting fuller answers. It has redundancy (“use only the text” vs “no outside knowledge”) and unnecessary tone constraints.

Prompt Refinement:

Original prompt: - Answer the question based only on the given text.

Refined prompt:

You are an AI assistant.

Answer the question using only the provided text.

Text: [Same paragraph]

Question: What is the main objective of football?

Instructions:

- Respond in 1–2 clear sentences
- Base your answer strictly on the given text
- Do not add outside information or assumptions
- If the answer is not stated, write "Not mentioned"

Tone: Formal and precise. Maintain a formal and precise tone.

Before vs after output comparison:

Before (Original Prompt Output):

The main objective of football is to score goals by getting the ball into the opposing team's net. Players must do this without using their hands, except for the goalkeeper.

After (Refined Prompt Output):

The objective of football is to score goals by getting the ball into the opposing team's net without using hands, except for the goalkeeper.

Prompt 2 : “You are an AI assistant.

Task: Answer the question based only on the given text.

Text:[Same paragraph]

Question:What challenges are associated with football??

Instructions:

- Answer clearly in 2-3 lines
- Use only the given text
- Do not include outside knowledge
- If answer is unclear, say "Not mentioned"

Tone: Simple, clear, friendly

Safety: No assumptions”

Sr. no	Output from ---	Output from ---	Analysis
2.	Football has challenges like player injuries, match-fixing scandals, and high competition pressure.	While football is super exciting, it does have some hurdles! The main challenges are player injuries, scandals like match-fixing, and the heavy pressure of competition. Plus, some people find that new tech like VAR can break up the natural flow of the match.	First answer: Clear, short, and strictly based on the text. Second answer: More engaging and detailed, but slightly less strict.

Prompt Failures & Biases Identified:Slightly restrictive and repetitive instructions; limits flexibility.

Prompt Refinement:

- **Original prompt:**Answer clearly in 2-3 lines using the text.
- **Refined prompt:**

You are a strict AI assistant.

Task: Identify challenges mentioned in the text.

Instructions:

- List only challenges explicitly given
- Do not add outside knowledge
- Answer in bullet points
- If none found, say "Not mentioned"

Output Format:

- Point 1
- Point 2

- **Before vs after output comparison:**

Before: Slightly longer, repetitive wording.

After: More concise and direct answer.

Prompt 3 :

You are an AI assistant.

Task: Answer the question based only on the given text.

Text: [Same paragraph]

Question: How does technology impact football?

Instructions:

- Answer based only on the text
- Do not give personal opinions
- If the answer is not directly stated, say "Cannot be determined from the text"

Tone: Analytical and slightly explanatory.

Sr. no	Output from ---	Output from ---	Analysis
3.	Technology like VAR helps improve fairness in decision-making. However, it also leads to debates as some believe it disrupts the natural flow of the game.	To analyze the role of technology within the sport, we can look at how modern tools attempt to balance competitive integrity with the traditional pace of the game	First answer: Direct, clear, and text-based. Second answer: More analytical but less direct.

Prompt Failures & Biases Identified:Instructions are slightly repetitive and restrict flexibility; tone requirement adds little value.

Prompt Refinement:

- **Original prompt:** Answer based on text
- **Refined prompt:**
You are an AI assistant tasked with extracting information strictly from the given text.
Input Text: [Same paragraph]
Query: How does technology impact football?
Response Guidelines:
 - Derive the answer only from explicitly stated information in the text
 - Keep the response brief (1–2 sentences) but include both positive and negative impacts if mentioned
 - Do not infer, assume, or add external knowledge
 - If the text does not clearly address the question, respond with: "Cannot be determined from the text"

Style Requirement:

Maintain an analytical tone with slight explanation, focusing on cause–effect relationships.

- **Before vs after output comparison:**
Before: Slightly wordy and constrained.
After: More concise, clear, and focused.

Prompt 4 :

You are an AI assistant.

Task: Answer the question based only on the given text.

Text:[Same paragraph]

Question:Does technology negatively affect football?

Instructions:

- Check if the statement is correct based on the text
- If incorrect, clearly explain why using the text
- Do not add extra knowledge

Tone: Critical

Safety: Stick strictly to given information

Sr. no	Output from Chatgpt	Output from Gemini	Analysis
4.	Technology like VAR may improve fairness, but it is criticized for disrupting the natural flow of the game. This creates ongoing debate about its overall impact.	While technology such as the Video Assistant Referee (VAR) is marketed as a tool for fairness, its implementation remains highly contentious. Based on the provided text, the impact of technology is not strictly positive	The first is precise and compliant with instructions, while the second is more critical but less strictly text-based.

Prompt Failures & Biases Identified:Encourages overly strict answers, limiting analysis. Slight bias toward neutrality, restricting critical tone.

Prompt Refinement:

- **Original prompt:** Answer using only the given text in 1–2 sentences.
- **Refined prompt:**
You are an AI assistant performing strict text-based evaluation.

Given Text: [Same paragraph]

Claim to Evaluate: Does technology negatively affect football?

Response Rules:

- Determine whether the claim is fully supported, partially supported, or not supported by the text
- Justify your judgment using only explicit evidence from the text
- Do not introduce assumptions, opinions, or outside knowledge
- Keep the response concise (1–2 sentences)

Tone Requirement:

Maintain a critical and cautious tone, clearly highlighting any limitations or ambiguity in the text.

- **Before vs after output comparison:** Before: Strict, direct, limited depth.
After: Still concise but slightly more analytical and insightful.

3.Role-based Prompt

Define Role based prompt: A role-based prompt is a prompt where the user assigns a specific role or persona (such as teacher, analyst, or journalist) to the AI to guide how it responds. By defining a role, the output becomes more structured, context-aware, and aligned with the expected tone, style, and expertise. This helps improve clarity, relevance, and consistency in the response.

Prompt 1 :

You are an AI assistant acting as a Sports Tutor.

Task: Answer the question based only on the given text.

Text:

Football is one of the most popular sports in the world, played by two teams of eleven players each. The objective of the game is to score goals by getting the ball into the opposing team's net without using hands, except for the goalkeeper. It requires a combination of physical fitness, teamwork, strategy, and skill. Major international tournaments like the FIFA World Cup attract millions of viewers globally. Football has a significant impact beyond entertainment, promoting unity and national pride among fans. However, it also has challenges such as player injuries, match-fixing scandals, and intense competition pressure. Advances in technology, like Video Assistant Referee (VAR), have been introduced to improve fairness in decision-making. Despite these developments, debates continue about whether technology disrupts the natural flow of the game.

Question:

What is the main objective of football?

Instructions:

- Answer in simple and easy language
- Keep the answer within 2–3 lines
- Explain as if teaching a beginner

Tone:

Friendly and easy to understand

Safety:

Do not include any information outside the given text.

Output from ChatGPT	Output from Gemini	Analysis
The main objective of football is to score goals by getting the ball into the opposing team's net. Players must do this without using their hands, except for the goalkeeper.	The main objective of football is to score goals by getting the ball into the opposing team's net. Except for the goalkeeper, players must do this without using their hands!	Both outputs are accurate and relevant to the given text. ChatGPT provides a more formal and instruction-following response, while Gemini offers a slightly more engaging but less formal answer. Overall, ChatGPT performs better for academic purposes, whereas Gemini is

		better for readability and engagement.
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Prompt Failures & Biases Identified:

Prompt Refinement:

Original prompt: You are an AI assistant.

Task: Answer the question based only on the given text.

Text:

[Same paragraph]

Question:

What is the main objective of football?

Instructions:

- Answer in simple and easy language
- Keep the answer within 2–3 lines

Tone:

Friendly and easy to understand

Safety:

Do not include any information outside the given text.

Refined prompt:

You are an AI assistant acting as a Sports Tutor.

Task: Answer the question based only on the given text.

Text:

[Same paragraph]

Question:

What is the main objective of football?

Instructions:

- Answer in exactly 2 lines
- Use clear and simple language
- Include both the goal and restriction (no hands except goalkeeper)
- Do not use expressive punctuation (e.g., !)
- Do not add or omit key details

Tone:

Formal yet easy to understand

Safety:

Strictly use only the given text. Avoid assumptions, exaggeration, or stylistic variation.

Before vs after output comparison:

The original prompt produced accurate but slightly inconsistent outputs, with variations in tone and structure (e.g., informal punctuation in one response). It lacked strict constraints, leading to minor deviations from the intended style. After refinement, the prompt enforced clear rules on format, tone, and content, resulting in more consistent and precise answers. Overall, the refined prompt improved clarity, uniformity, and adherence to instructions.

Prompt 2 :

You are an AI assistant acting as an Academic Expert.

Task: Answer the question based only on the given text.

Text: Football is one of the most popular sports in the world, played by two teams of eleven players each.

The objective of the game is to score goals by getting the ball into the opposing team's net without using hands, except for the goalkeeper. It requires a combination of physical fitness, teamwork, strategy, and skill. Major international tournaments like the FIFA World Cup attract millions of viewers globally.

Football has a significant impact beyond entertainment, promoting unity and national pride among fans.

However, it also has challenges such as player injuries, match-fixing scandals, and intense competition pressure. Advances in technology, like Video Assistant Referee (VAR), have been introduced to improve fairness in decision-making. Despite these developments, debates continue about whether technology disrupts the natural flow of the game.

Question:

What is the fundamental objective of football?

Instructions:

- Use formal and precise language
- Provide a clear and concise explanation
- Limit the answer to 3–4 lines

Tone:

Formal and academic

Safety:

Ensure accuracy and avoid adding external information.

Sr. no	Output from chatgpt	Output from gemini	Analysis
1	The fundamental objective of football is to score goals by successfully placing the ball into the opposing team's net. This must be achieved without the use of hands, except by the goalkeeper. The objective defines the core rule and structure of the game.	The fundamental objective of football is to score goals by successfully maneuvering the ball into the opposing team's net. This must be achieved without the use of hands by any participant, with the sole exception of the goalkeeper. Accomplishing this goal necessitates a strategic integration of physical fitness, collective teamwork, and technical skill.	Both outputs are accurate and aligned with the given text, correctly explaining the fundamental objective of football. The ChatGPT response is more concise and strictly follows the core information without adding extra details, making it better for instruction-based tasks. In contrast, the Gemini response is more elaborate and includes additional elements like "strategy, teamwork, and technical skill," which were not directly required for the question. This shows a slight tendency toward over-elaboration in Gemini, whereas ChatGPT

			demonstrates better instruction adherence and precision.
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Prompt Failures & Biases Identified:

Over-elaboration: Gemini added extra details (skills, teamwork) not required.

Instruction drift: Response went beyond the specific question.

Prompt Refinement:

Original prompt: Explain the fundamental objective of football.

Refined prompt: You are an AI assistant.

Task: Answer the question based only on the given text.

Question:

What is the fundamental objective of football?

Instructions:

- Answer in 2–3 lines only
- Focus strictly on the objective
- Do not include skills or extra details

Tone:

Formal and precise

Safety:

Do not add information beyond the given text.

Before vs after output comparison: Before refinement, the response included unnecessary details beyond the main objective, leading to slight deviation from the question. After refinement, the output became more focused, concise, and strictly aligned with the required information. This improved accuracy and ensured better adherence to instructions.

Prompt 3 :

You are an AI assistant acting as a Sports Analyst.

Task: Answer the question based only on the given text.

Text:

[Same paragraph]

Question:

What challenges are associated with football?

Instructions:

- Answer in bullet points
- Highlight key issues only
- Keep it concise

Tone:

Analytical and concise

Safety:

Do not assume or include information beyond the text.

Sr. no	Output from chatgpt	Output from gemini	Analysis
1	● Player injuries ● Match-fixing scandals ● Intense competition pressure ● Concerns over technology (e.g., VAR) disrupting the natural flow of the game	Based on the provided text, the challenges associated with football include: ● Player injuries: Physical risks and health setbacks faced by athletes. ● Match-fixing scandals: Ethical issues and corruption regarding the integrity of the game. ● Intense competition pressure: The psychological and performance-based stress on participants. ● Technological debates: Concerns regarding whether advancements like VAR disrupt the natural flow of the game.	Both outputs correctly identify the challenges associated with football as mentioned in the text. The ChatGPT response is concise and presented in a simple list format, making it easy to read and directly aligned with the instructions. On the other hand, the Gemini response provides a more detailed explanation for each point, improving clarity but slightly increasing length. This shows that ChatGPT performs better in brevity and strict instruction adherence, while Gemini demonstrates better elaboration and interpretability.

Prompt Failures & Biases Identified:

Variation in format: ChatGPT gave only points, while Gemini added explanations

Instruction looseness: Prompt did not clearly specify whether explanation was required

Inconsistency: Different output styles despite same input

Prompt Refinement:

- **Original prompt:**What challenges are associated with football?
- **Refined prompt:**

You are an AI assistant.

Task: Answer the question based only on the given text.

Question:

What challenges are associated with football?

Instructions:

- List only the key challenges
- Use bullet points
- Do not provide explanations
- Keep answer concise

Tone:

Clear and structured

Safety:

Do not include information outside the given text.

- **Before vs after output comparison:** Before refinement, outputs varied in format, with some including explanations and others listing only points. After refinement, the responses became consistent, strictly formatted as bullet points, and more aligned with the instructions, improving clarity and uniformity.

Prompt 4 :

You are an AI assistant acting as a Critical Thinker.

Task: Answer the question based only on the given text.

Text:

[Same paragraph]

Question:

How does technology like VAR impact football?

Instructions:

- Provide a balanced explanation
- Base your answer strictly on the text
- Keep the answer structured in 3–4 lines

Tone:

Thoughtful and balanced

Safety:

Avoid bias and do not introduce personal opinions or external facts.

Sr. no	Output from chatgpt	Output from gemini	Analysis
1	Technology such as Video Assistant Referee (VAR) has enhanced football by improving fairness and accuracy in	The introduction of Video Assistant Referee (VAR) is intended to enhance the game by improving fairness in decision-making. However, its impact is a subject of ongoing	If you want a quick, scannable summary, the first output is effective. However, for a Critical Thinker task, the second output is superior because it

	decision-making. It helps reduce errors in critical situations during matches. However, it has also led to concerns about interrupting the natural flow of the game. As a result, debates continue regarding its overall impact on the sport.	debate, as it raises concerns about potentially disrupting the natural flow of the match. Consequently, technology in football serves as both a tool for accuracy and a point of contention regarding gameplay rhythm.	synthesizes the tension between technology and tradition into a cohesive argument.
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Prompt Failures & Biases Identified:

- Tone inconsistency: Some outputs became slightly opinion-based instead of balanced
- Ambiguity: Prompt did not clearly restrict interpretation of “impact”
- Over-interpretation: Model may add pros/cons not strictly mentioned in the text

Prompt Refinement:

- **Original prompt:**How does technology like VAR impact football?

Refined prompt:You are an AI assistant.

Task: Answer the question based only on the given text.

Question:

How does technology like VAR impact football?

Instructions:

- Provide a balanced answer (positive and concern)
- Limit response to 3–4 lines
- Do not include opinions beyond the text

Tone:

Formal and neutral

Safety:

Avoid bias and do not add external information.

Before vs after output comparison:

Before refinement, the responses sometimes included extra opinions or unclear interpretations of VAR's impact. After refinement, the answers became more balanced, precise, and strictly based on the given text, improving consistency and reducing bias.

1. json-based prompts

Define json Based Prompt: A JSON-based prompt is a structured way of giving instructions to a language model using JSON (JavaScript Object Notation) format, where inputs like text, question, constraints, and output format are clearly defined as key-value pairs.

Prompt 1 :

Prompt statement- Answer a direct factual question from the given text.

json format prompt-

```
{"role": "AI assistant",  
  
"task": "question_answering",  
  
"text": "Football is one of the most popular sports in the world... (same paragraph)",  
  
"question": "What is the objective of football?",  
  
"instructions": {  
  
  "answer_length": "1-2 lines",  
  
  "use_only_text": true,  
  
  "no_extra_information": true,  
  
  "fallback": "Not mentioned"  
  
},
```

```
"output_format": {

  "answer": "string"

}}
```

Sr. no	Output from ChatGPT	Output from Gemini	Analysis
1.	{ "answer": "The objective of football is to score goals by getting the ball into the opposing team's net without using hands, except for the goalkeeper." }	{ "answer": "The objective of the game is to score goals by getting the ball into the opposing team's net without using hands, except for the goalkeeper." }	Both models show high accuracy and clarity for straightforward factual questions, with no hallucination or ambiguity.

Prompt Failures & Biases Identified:

Over-generalization: Model may simplify answer to “to win the game” instead of exact objective.

Minor Hallucination: Sometimes adds extra phrases like teamwork/strategy which are not part of the objective.

Prompt Refinement:

- **Original prompt: Basic JSON prompt without strict constraints on precision.**
- **Refined prompt:**

```
{
  "role": "factual AI assistant",
  "task": "question_answering",
  "text": "Football paragraph",
  "question": "What is the objective of football?",
  "instructions": {
    "answer_length": "exact 1 sentence",
    "focus": "only objective statement",
    "no_extra_information": true,
    "strict_extraction": true,
    "fallback": "Not mentioned"
  },
  "output_format": {
```

```
"answer": "string"
}}
```

- **Before vs after output comparison:**

Before:

“The objective of football is to score goals and win the game using teamwork.”

After:

“The objective of football is to score goals by getting the ball into the opposing team’s net without using hands, except for the goalkeeper.”

Prompt 2 :

Prompt statement- Identify challenges mentioned in the text.

json format prompt-

```
{ "role": "AI assistant",

"task": "information_extraction",

"text": "Football is one of the most popular sports in the world... (same paragraph)",

"question": "What challenges are associated with football?",

"instructions": {

"extract_only_listed_items": true,

"no_external_knowledge": true,

"format": "bullet_points"

},

"output_format": {

"challenges": ["string"]

}}
```

Sr. no	Output from ---	Output from ---	Analysis
2.	{ "challenges": [	{ "challenges": [	Both models

	"Player injuries", "Match-fixing scandals", "Intense competition pressure", "Debates about whether technology disrupts the natural flow of the game"] }	"player injuries", "match-fixing scandals", "intense competition pressure", "debates about technology disrupting the natural flow of the game"] }	perform well in information extraction, with ChatGPT having slightly better presentation clarity.
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Prompt Failures & Biases Identified:

Hallucination: Model may add challenges not mentioned (e.g., financial issues).

Over-generalization: Might summarize instead of listing exact points.

Prompt Refinement:

- **Original prompt: JSON prompt asking for challenges without strict extraction rules.**

- **Refined prompt:**

```
{
  "role": "strict extraction AI",
  "task": "information_extraction",
  "text": "Football paragraph",
  "question": "What challenges are associated with football?",
  "instructions": {
    "extract_only_explicit_items": true,
    "no_inference": true,
    "no_external_knowledge": true,
    "preserve_text_phrases": true
  },
  "output_format": {
    "challenges": ["exact phrases from text"]
  }
}
```

- **Before vs after output comparison:**

Before:

“Football faces injuries, financial problems, and pressure.”

After:

Player injuries

Match-fixing scandals
Intense competition pressure

Prompt 3 :

Prompt statement- Provide a balanced answer for a subjective question.

json format prompt-

```
{ "role": "neutral AI assistant",  
  
  "task": "analysis",  
  
  "text": "Football is one of the most popular sports in the world... (same paragraph)",  
  
  "question": "Is technology good for football?",  
  
  "instructions": {  
  
    "no_personal_opinion": true,  
  
    "use_only_given_text": true,  
  
    "include_pros_and_cons": true,  
  
    "fallback": "Cannot be determined from the text"  
  
  },  
  
  "output_format": {  
  
    "analysis": "string"  
  
  }  
}
```

Sr. no	Output from ---	Output from ---	Analysis
3.	{ "analysis": "The text indicates that technology, such as VAR, has been introduced to improve fairness in decision-making, which is a	{ "analysis": "Based on the text, technology presents both benefits and drawbacks. It is described as an advancement	Both models show strong capability in handling ambiguous questions, providing neutral

	positive aspect. However, it also mentions ongoing debates about whether technology disrupts the natural flow of the game, which is a negative aspect. Therefore, both advantages and concerns are presented, and no definitive conclusion is given." }	intended to improve fairness in decision-making through tools like the Video Assistant Referee (VAR); however, it also generates debate as it is perceived by some to disrupt the natural flow of the game." }	and balanced analysis.
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Prompt Failures & Biases Identified:

Ambiguity: Model may give one-sided answer.

Bias: May favor a positive view of technology.

Prompt Refinement:

- **Original prompt: JSON prompt asking for analysis without neutrality constraint.**

- **Refined prompt:**

```
{
  "role": "neutral analysis assistant",
  "task": "balanced_reasoning",
  "text": "Football paragraph",
  "question": "Is technology good for football?",
  "instructions": {
    "include_both_sides": true,
    "no_personal_opinion": true,
    "strict_text_basis": true,
    "avoid_conclusion_bias": true,
    "fallback": "Cannot be determined from the text"
  },
  "output_format": {
    "analysis": "balanced explanation with pros and cons"
  }
}
```

- **Before vs after output comparison:**

Before:

“Technology is good because it improves fairness.”

After:

“Technology improves fairness through VAR, but it may disrupt the natural flow of the game, so both positive and negative aspects exist.”

Prompt 4 :

Prompt statement- Verify correctness of a given statement using the text.

json format prompt-

```
{ "role": "logical AI assistant",  
  
  "task": "fact_verification",  
  
  "text": "Football is one of the most popular sports in the world... (same paragraph)",  
  
  "question": "Football has no negative aspects. Explain.",  
  
  "instructions": {  
  
    "verify_statement": true,  
  
    "reject_if_false": true,  
  
    "use_text_evidence": true,  
  
    "no_assumptions": true  
  
  },  
  
  "output_format": {  
  
    "verdict": "Correct/Incorrect",  
  
    "explanation": "string"  
  
  }  
}
```

Sr. no	Output from ChatGPT	Output from Gemini	Analysis
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4.	<pre>{ "verdict": "Incorrect", "explanation": "The text explicitly mentions negative aspects such as player injuries, match-fixing scandals, and intense competition pressure. It also refers to debates about technology disrupting the natural flow of the game, showing that football does have negative aspects." }</pre>	<pre>{ "verdict": "Incorrect", "explanation": "The text explicitly identifies several negative aspects of football, including player injuries, match-fixing scandals, and intense competition pressure. It also notes that technological developments like VAR have introduced debates regarding the disruption of the game's natural flow." }</pre>	Both models are effective in fact verification tasks, correctly rejecting false statements and supporting answers with textual evidence.
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Prompt Failures & Biases Identified:

Incorrect Assumption: Model may accept false statement.

Hallucination: May justify incorrect claim.

Prompt Refinement:

- **Original prompt: JSON prompt asking for explanation without verification step.**
- **Refined prompt:**

```
{
  "role": "logical verification assistant",
  "task": "fact_checking",
  "text": "Football paragraph",
  "question": "Football has no negative aspects. Explain.",
  "instructions": {
    "verify_truth": true,
    "reject_false_statements": true,
    "provide_text_evidence": true,
    "no_assumptions": true
  },
  "output_format": {
    "verdict": "Correct/Incorrect",
    "evidence": ["points from text"],
    "explanation": "string"
  }
}
```

- **Before vs after output comparison:**

Before:

“Football mostly has positive aspects and brings unity.”

After:

Verdict: Incorrect

Evidence: Player injuries, match-fixing scandals, competition pressure

Explanation: The text clearly mentions negative aspects.

Discussion, takeaway

The experiment demonstrated that prompt design plays a crucial role in determining the quality of responses generated by LLMs. Different prompting techniques produced varying levels of accuracy, clarity, and completeness.

Zero-shot prompts were simple and efficient but sometimes lacked precision, especially for complex or ambiguous queries. Few-shot prompting improved response consistency by providing examples, helping the model better understand the expected format. Role-based prompting enhanced contextual relevance and tone, making responses more aligned with specific tasks such as tutoring or analysis. JSON-based prompts provided structured and consistent outputs, reducing ambiguity and improving reliability in information extraction tasks.

A key observation was that poorly defined prompts often led to hallucination, over-generalization, or incorrect assumptions. However, prompt refinement using clear constraints, structured formats, and explicit instructions significantly improved performance.

Overall, the study highlights the importance of carefully designing prompts to achieve accurate, unbiased, and context-aware outputs from AI systems.

Conclusion

This experiment successfully demonstrated how different prompt engineering techniques affect the behavior of Large Language Models in question-answering tasks. It was observed that structured and well-defined prompts lead to more accurate and reliable outputs, while vague prompts increase the chances of errors such as hallucination and ambiguity.

Among the techniques explored, refined prompts and structured formats like JSON-based prompting provided the most consistent results. The study also emphasized the importance of identifying and addressing biases and limitations in AI-generated responses.

In conclusion, prompt engineering is a powerful tool for improving LLM performance, and careful design, testing, and refinement of prompts are essential for responsible and effective AI usage.

Prompt Type	Best For	Key Weakness	Rating
Zero-Shot	Simple factual questions	Hallucination on contradictory inputs	★ ★ ★
Few-Shot	Consistent format; known patterns	May bias format over content fidelity	★ ★ ★ ★
Role-Based	Citation-backed academic answers	Persona introduces diplomatic over-generalisation	★ ★ ★ ★
JSON-Based	Downstream processing; traceability	Rigid schema struggles with ambiguity	★ ★ ★ ★
Chain-of-Thought	Complex, inferential, multi-hop questions	Verbose for simple queries	★ ★ ★ ★ ★

References

- OpenAI ChatGPT Documentation
- Google Gemini AI
- Course materials and lab instructions provided by the Department of Computer Engineering
- General knowledge text on football (used as dataset for experimentation)